

Rethinking Walkability: A Thermal Perspective for TOD and Beyond

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Outline

1. Concept
2. Measures
3. A Thermal Perspective of Walkability for TOD
4. Exploratory Analysis
 - 4.1 Arlington, VA
 - 4.2 Dallas, TX
 - 4.3 Austin, TX
5. Concluding Remarks

1. Concept

Defined by Dictionary and AI

Dictionary Definition:

The **state or quality** of being *walkable* (Dictionary.com)

Walkable: Capable of or suitable for being walked (Merriam-Webster)

ChatGPT Definition:

The extent to which an area is **safe, convenient, comfortable, and attractive** for people to travel on foot to reach everyday destinations.

1. Concept

Defined by Planners/Designers/Engineers

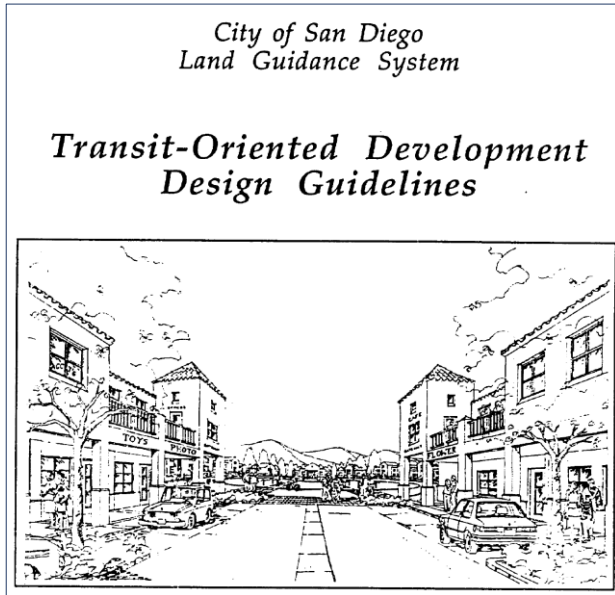
“Walkability is the extent to which the built environment supports and encourages walking by providing for pedestrian **comfort and safety**, **connecting people with varied destinations** within a reasonable amount of time and effort, and offering **visual interest** in journeys throughout the network.”

-Southworth, M. 2005. “Designing the Walkable City.” *Journal of Urban Planning and Development* 131 (4): 246–257.

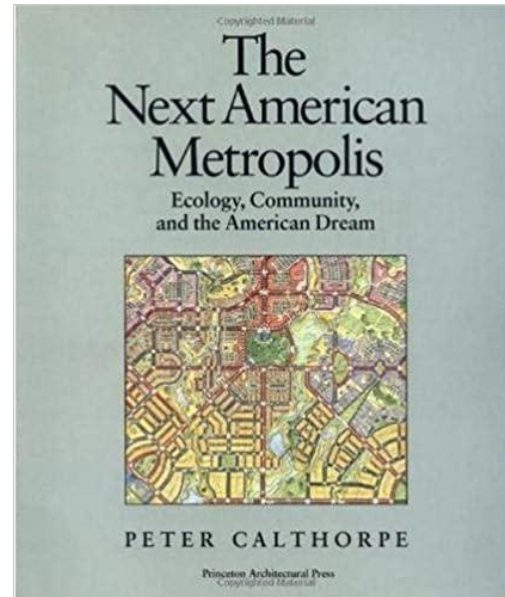
“Walkability refers to the overall **quality and ability of an environment** to promote and support pedestrian travel for a range of purposes.”

- Litman, T. 2014. Walkability. In A. C. Michalos (Ed.), *Encyclopedia of Quality of Life and Well-Being Research* (pp. 6993–6995).

1. Concept: Walkability An Essential Attribute of TOD



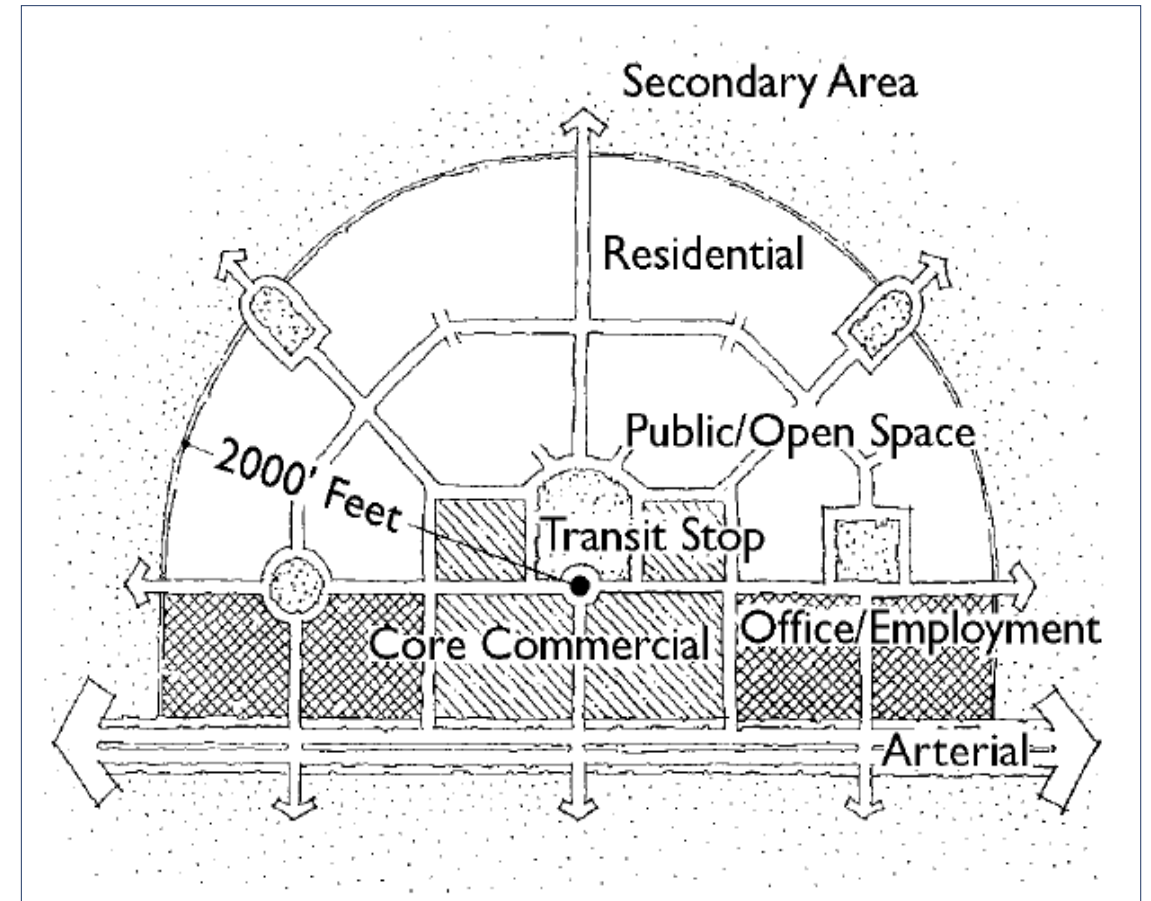
Calthorpe, 1992



Calthorpe, 1993

TOD (3-D Attributes)

- Higher Density (15 units/acre)
- Mixed Uses (Diversity)
- **Pedestrian Friendly Environment** (Design)
- 400-800 Meters Radius
- Transit Services



(Peter Calthorpe, 1993, The Next American Metropolis)

2. Measures: Simplified, Network-Based

San Juan, Puerto Rico Application

Paper No. 99-0892

TRANSPORTATION RESEARCH RECORD 1669

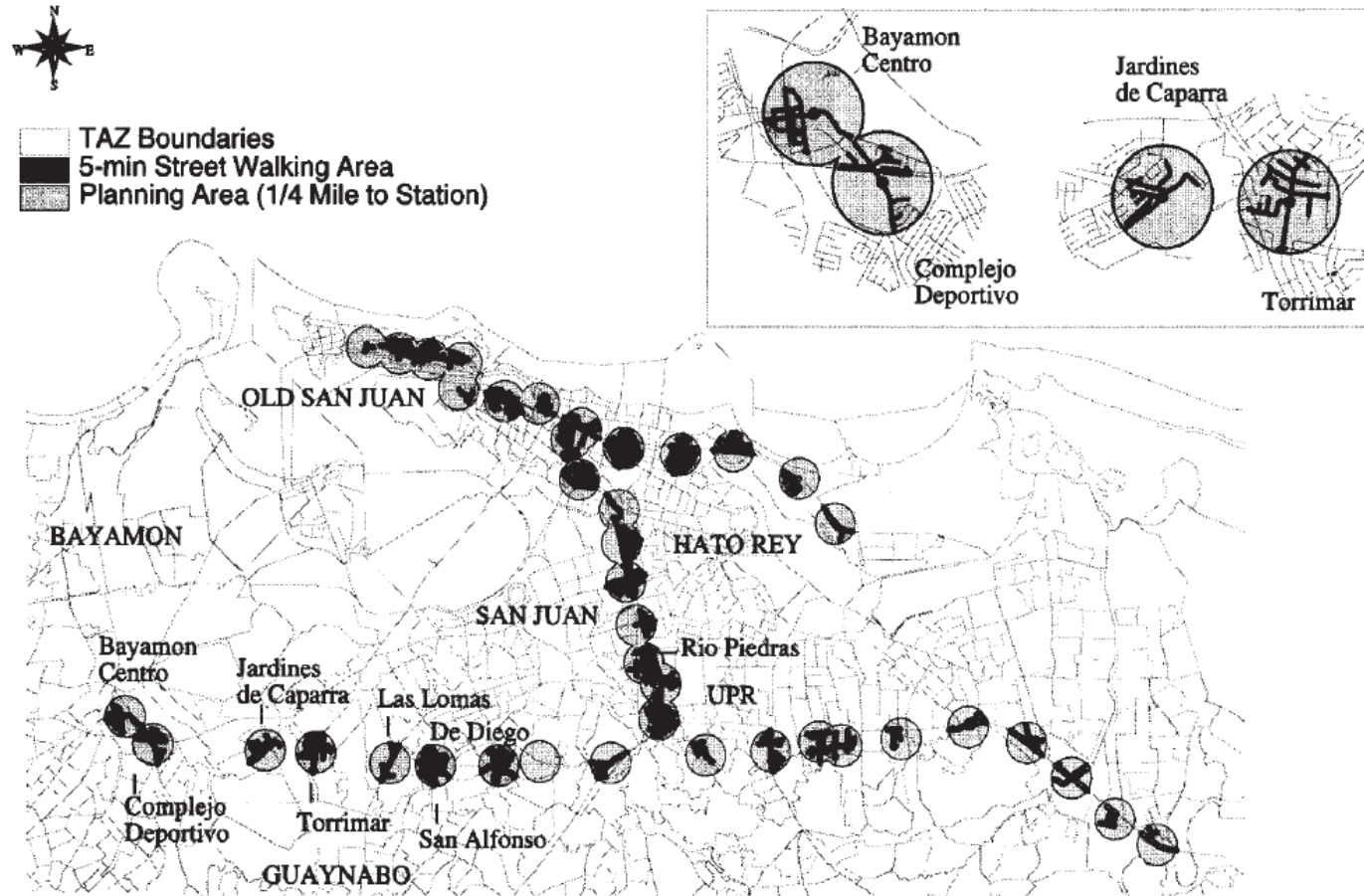


FIGURE 1 Evaluation of station area walking environment: example of Tren Urbano in the SJMR. 1 mile = 1.61 km.

Walkability Index (WI):
Areas reachable by walking network divided by the areas covered in airline distance or time

WI:

$$\frac{[2r^2 - (2r^2/n)]}{r^2 \cdot \pi}$$
$$= [2 - (2/n)] / \pi$$

where:
r: TOD radius
n: Network density

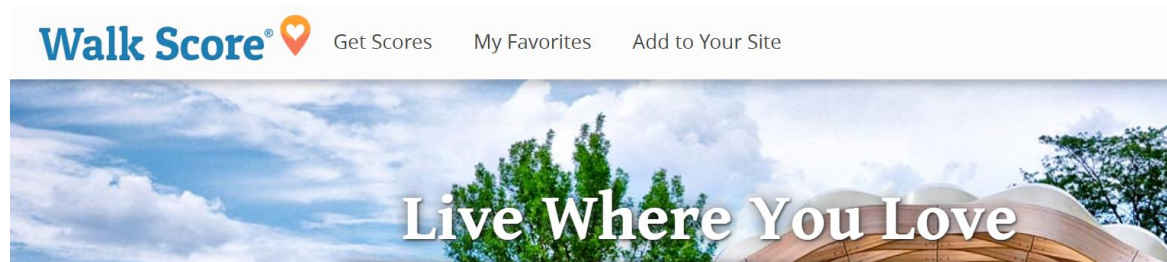
Zhang, et al. 1999

2. Measures: Composite, Comprehensive

- Jane Jacobs, 1960: Small blocks, mixed use, higher density
- Frank, L.D., et al., 2005. Linking objectively measured physical activity with objectively measured urban form: findings from SMARTRAQ. *American journal of preventive medicine*, 28(2), pp.117-125.

$$\begin{aligned}\text{Walkability index} = & (6 \times \text{z-score of land-use mix}) \\ & + (\text{z-score of net residential density}) \\ & + (\text{z-score of intersection density})\end{aligned}$$

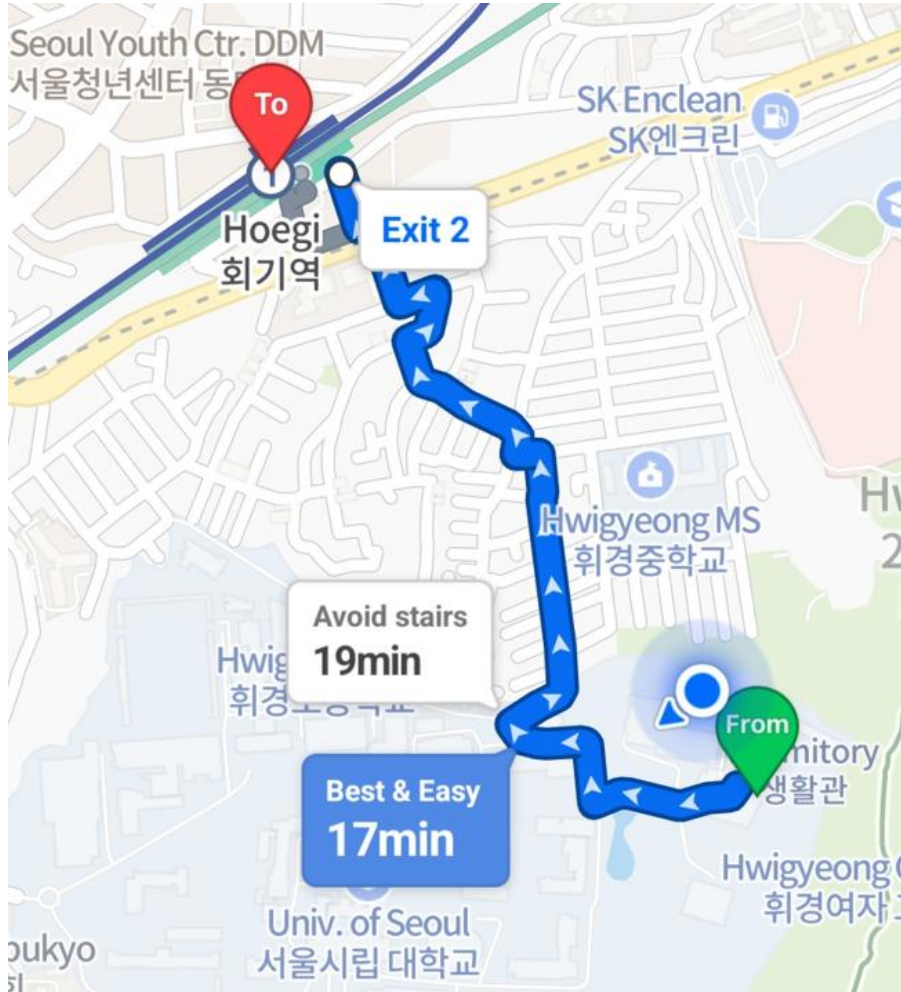
- Walk Score



City-wide average Walk Score: Seoul: 90; Austin: 42

$$\text{Walk Score} = 100 \times \frac{\sum_{i=1}^n w_i f(d_i)}{\sum_{i=1}^n w_i}$$

2. Measures: Hoegi-International Dorm: Walkable Route!?



Walkability deteriorates during hot times/days.



3. A Thermal Perspective of Walkability for TOD

Ongoing research by Do Yun Lee, Ph.D. student, UT Austin

Extreme heat can function as a barrier to mobility and accessibility, not just a comfort issue

- Microscale thermal modeling
- Pedestrian network analysis
- Thermally conditioned walkability/accessibility

3. A Thermal Perspective of Walkability for TOD

Universal Thermal Comfort Index

The Universal Thermal Climate Index, UTCI, is a bioclimatic index for describing the physiological comfort of the human body **under specific meteorological conditions** (Bröde et al. 2012)

It considers not just the ambient temperature but also **other variables (humidity, wind and radiation)**

More accurate indicator to use when analyzing the impacts that the environment conditions have on the human body and on people's behavior

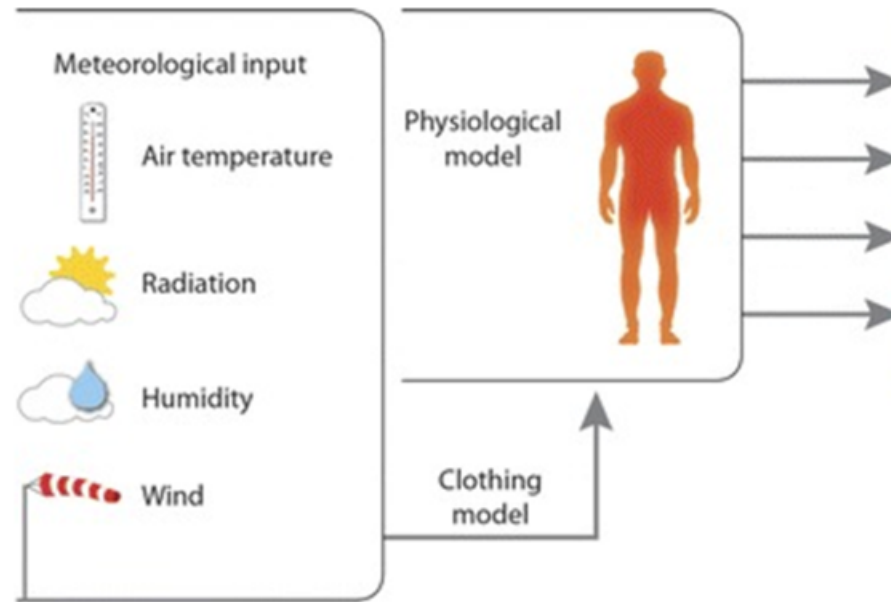
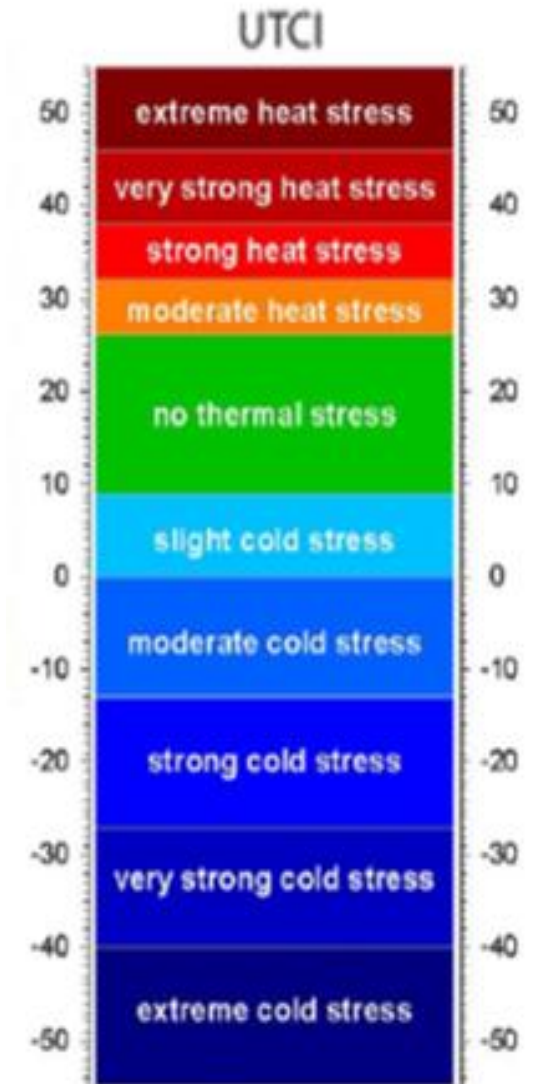


Figure 1. Diagram of the UTCI model



3. A Thermal Perspective of Walkability for TOD

Data

Spatial Inputs

- 1-m LiDAR-derived DEM, DSM, CDSM
- Sky View Factor (SVF), shading geometry

Meteorological Forcing

- ERA5 air temperature, relative humidity
- Incoming shortwave radiation

Transit Supply

- GTFS stop-level service frequency $S(s,t)$

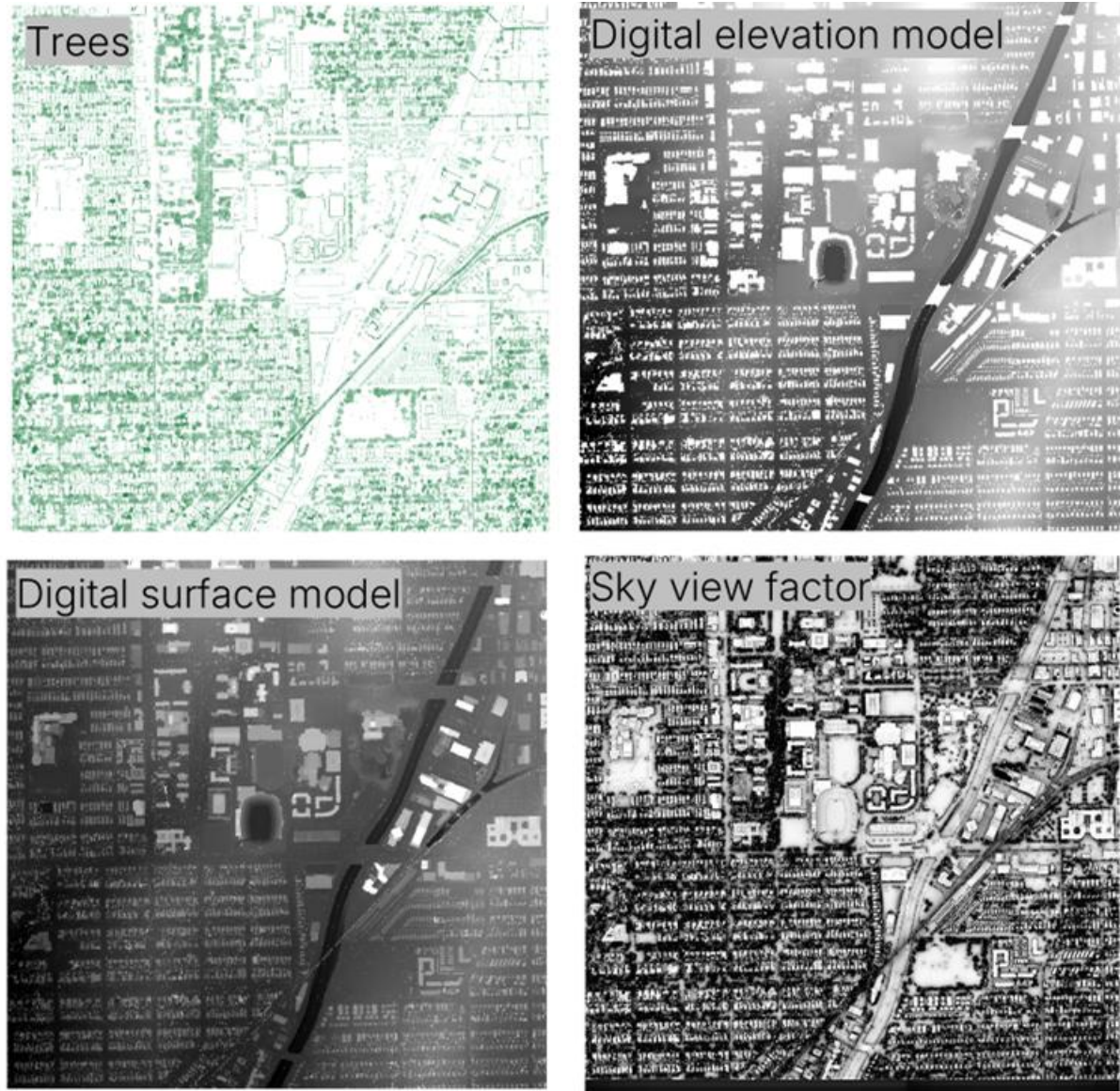
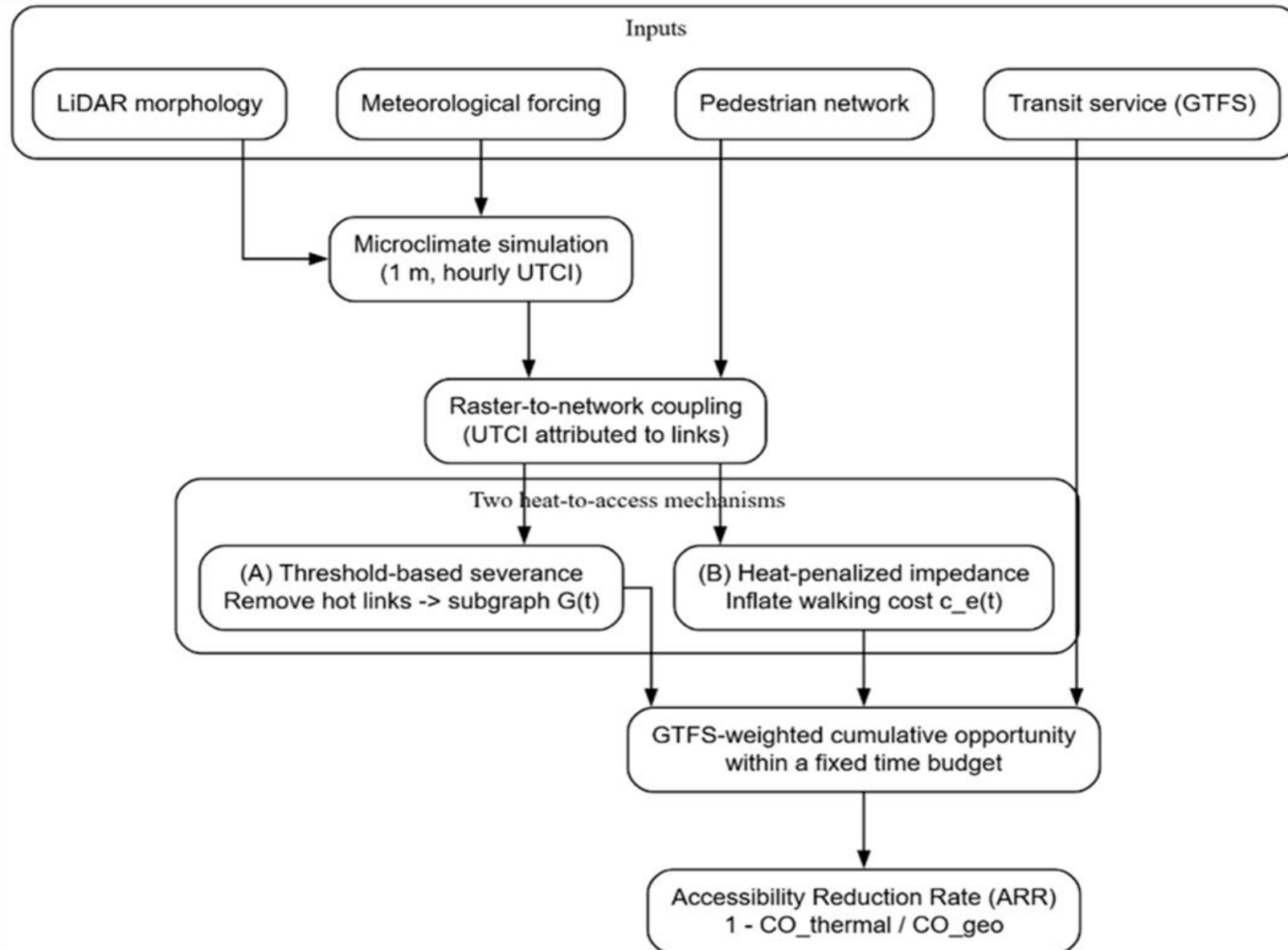


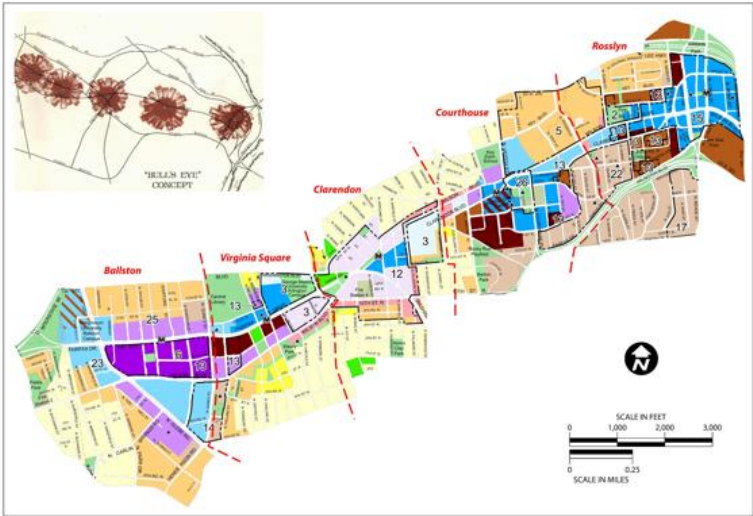
Figure 2. Lidar data visualization for Mockingbird station, Dallas

3. A Thermal Perspective of Walkability for TOD



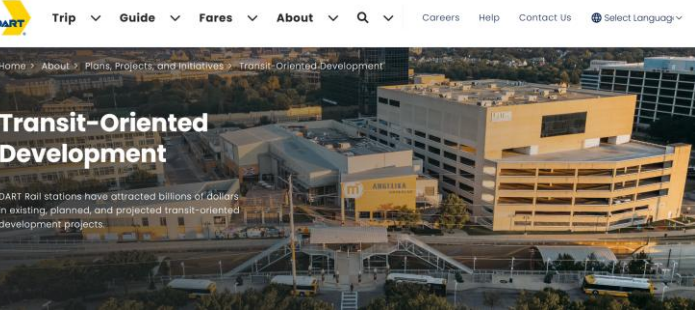
4. Exploratory Analysis: Three Cases

Arlington, VA



Dallas, TX

Rail System Map/ Mapa del sistema ferroviario



Austin, TX



4. Exploratory Analysis: Results

Thermal Exposure Is Highly Uneven Within Station Areas

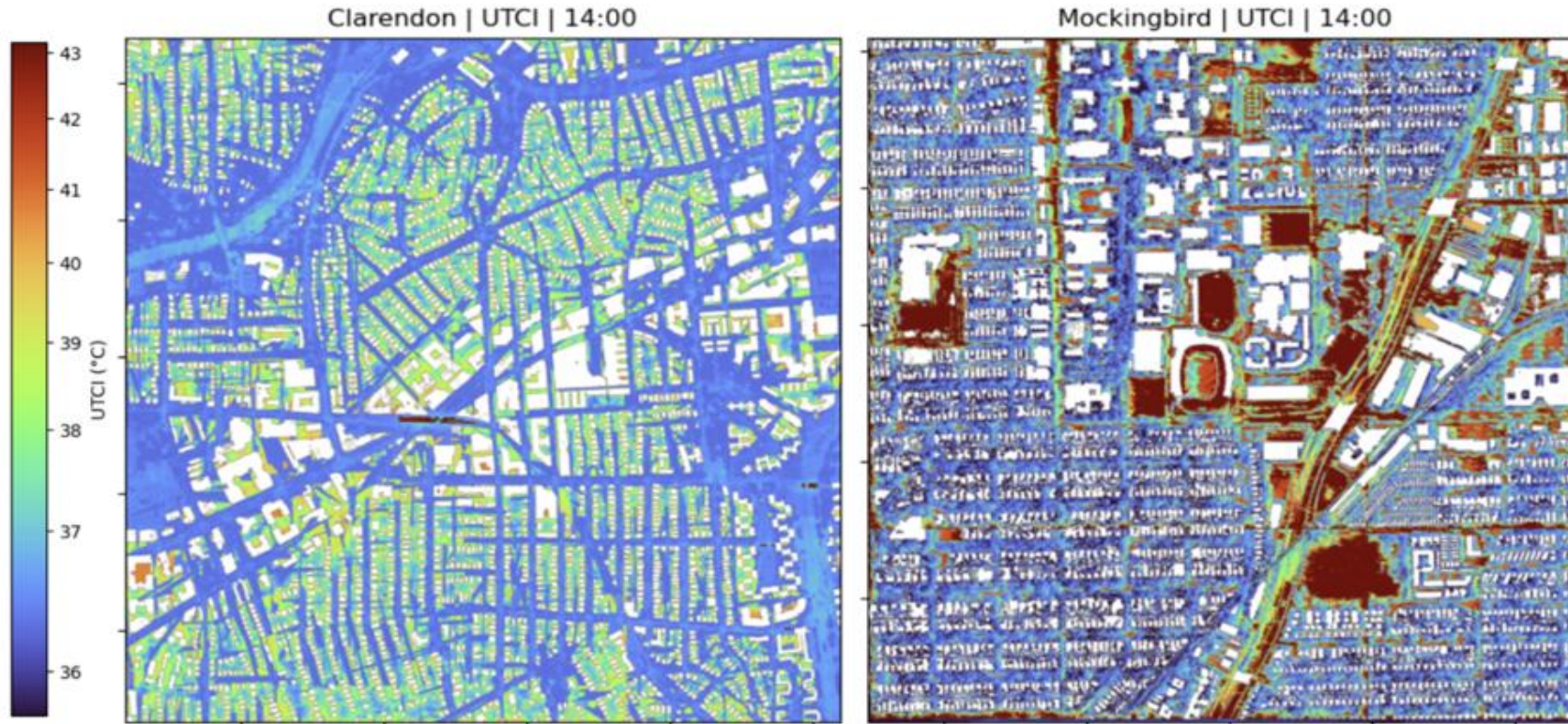


Figure 3. Spatial heterogeneity of 1-m UTCI at 14:00 in representative station areas (Clarendon, Arlington, VA and Mockingbird, Dallas, TX)

Thermal conditions vary block by block.

Shaded street canyons create cooler pedestrian corridors.

Exposed parcels and wide arterials form continuous high-stress zones.

4. Exploratory Analysis: Results

Urban Form Systematically Structures Pedestrian Heat Exposure

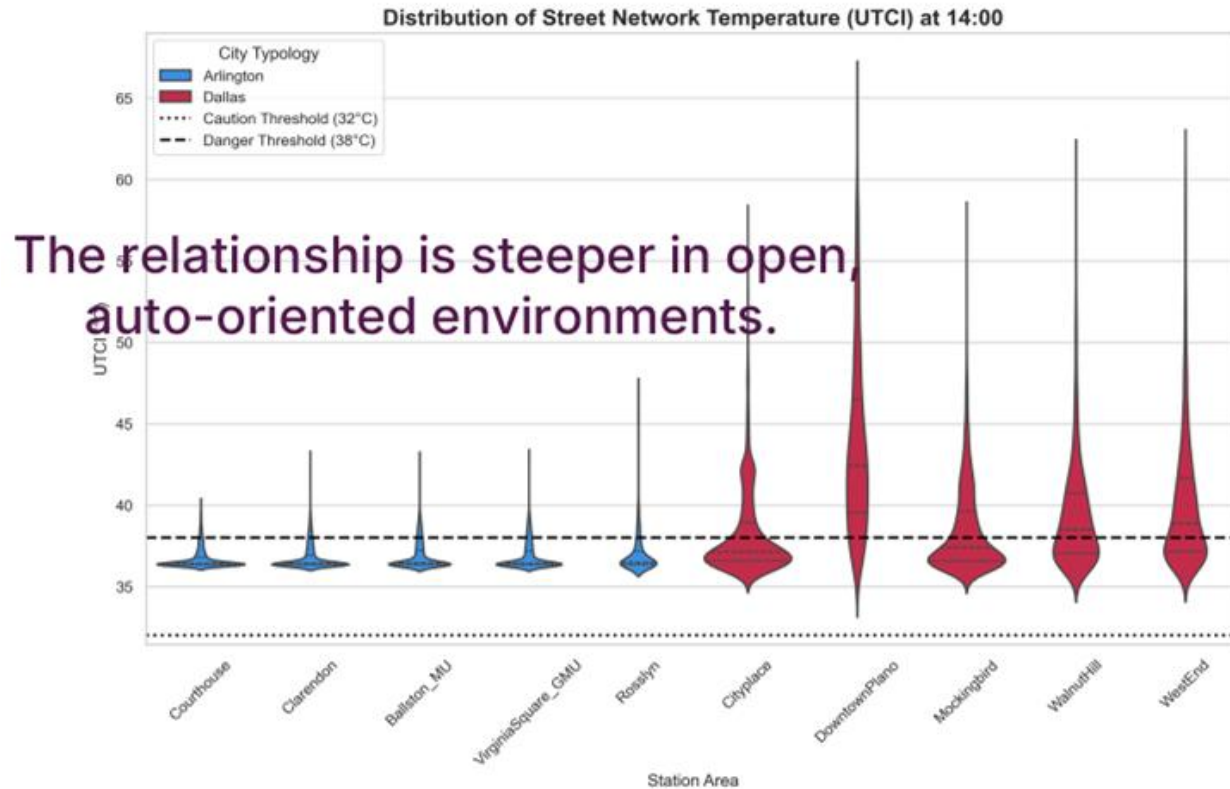


Figure 4a. Station-area UTCI distributions at 14:00 across Arlington and Dallas stations)

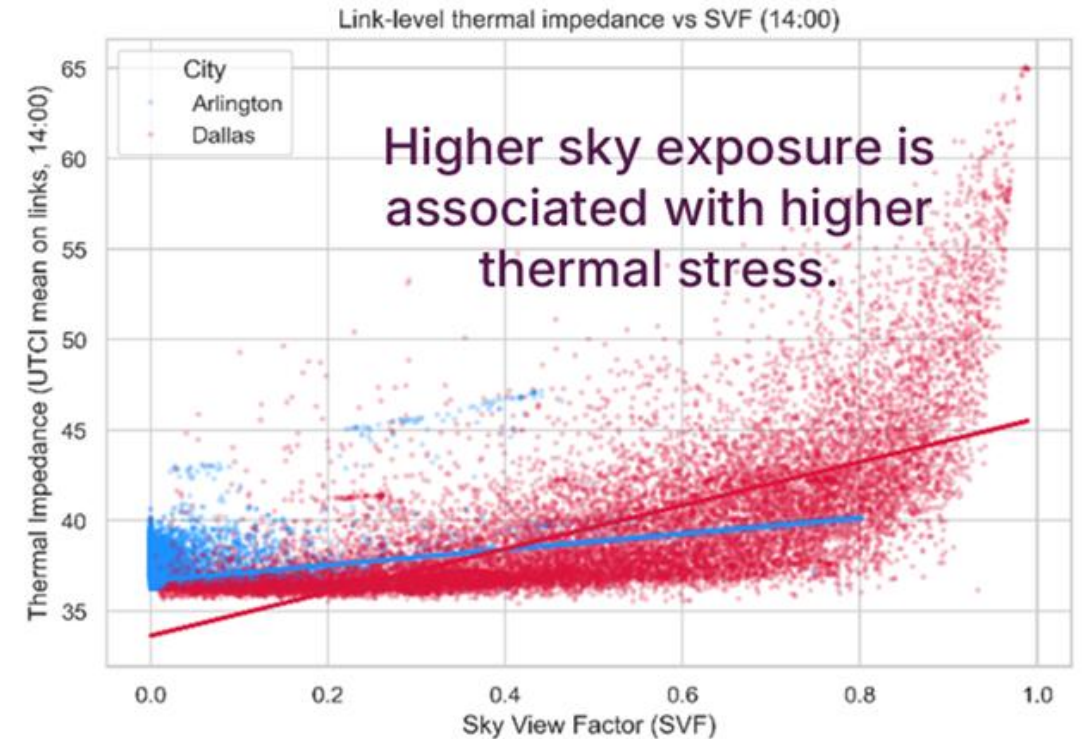
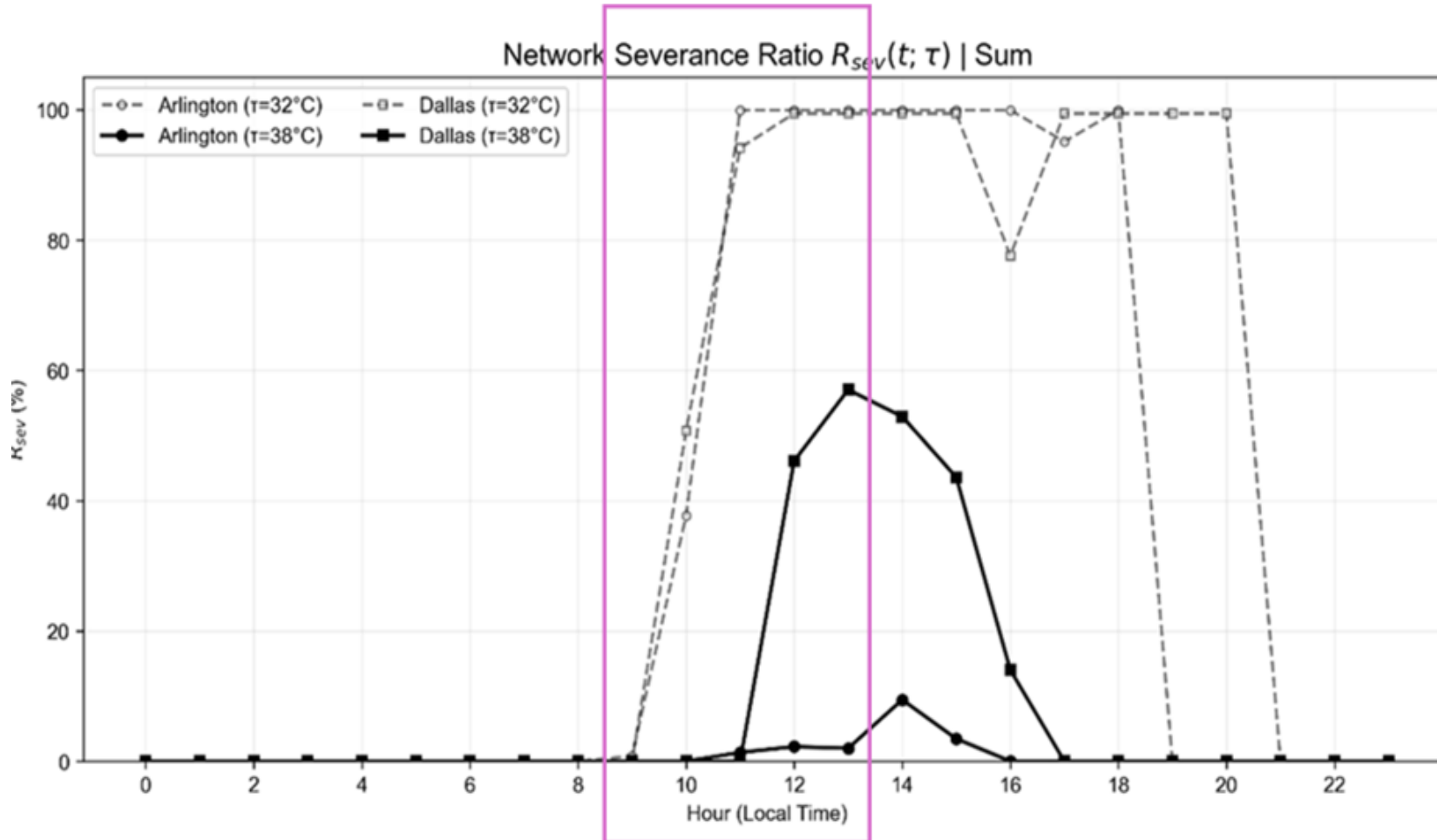


Figure 4b. Relationship between link-level thermal impedance (mean UTCI on links at 14:00) and Sky View Factor (SVF)

4. Exploratory Analysis: Results

Extreme Heat Produces Time-Specific Network Severance



Early afternoon shows sharp increases in network severance

Figure 5. Network severance ratio over time in Arlington and Dallas

4. Exploratory Analysis: Results

Extreme Heat Produces Time-Specific Network Severance

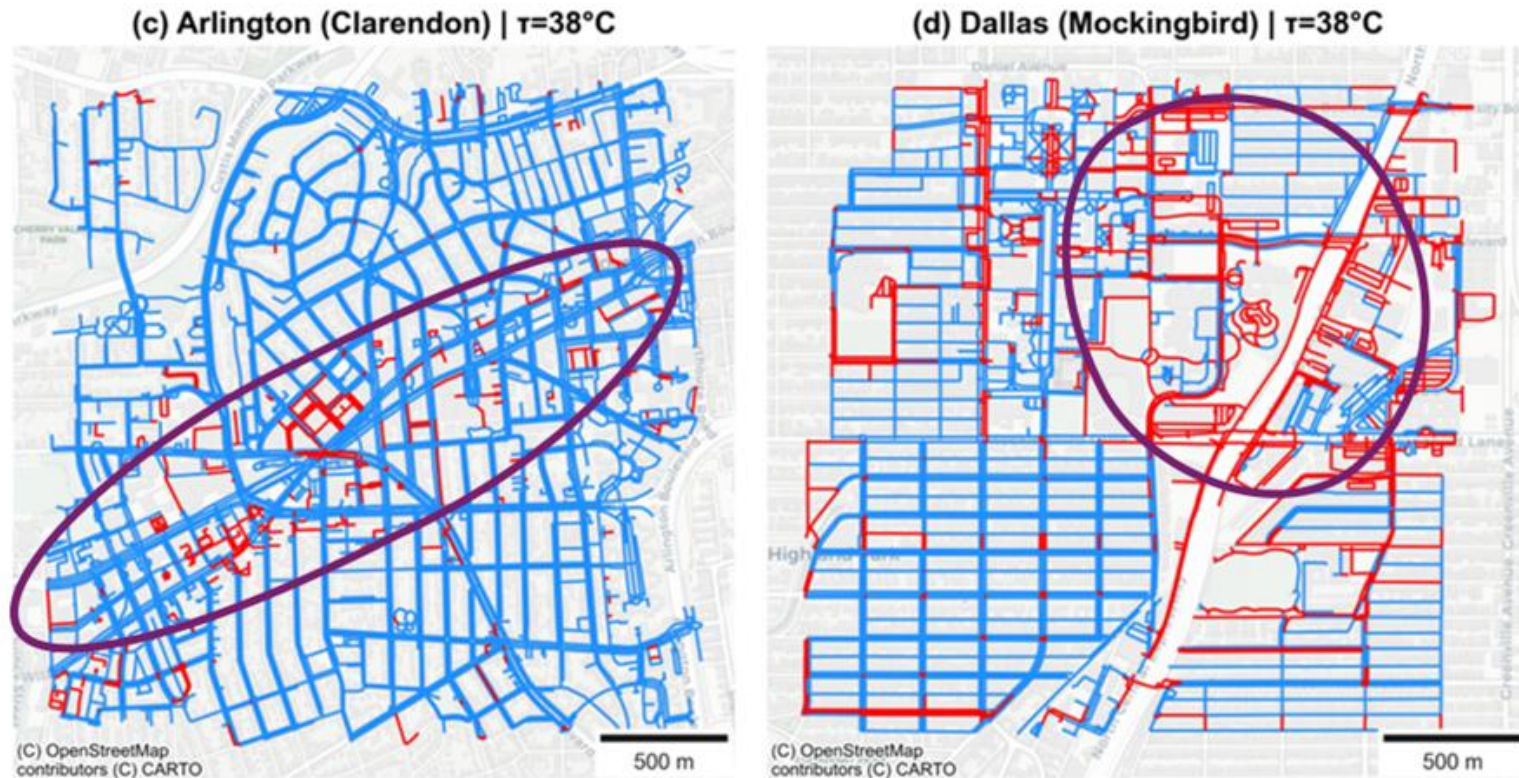


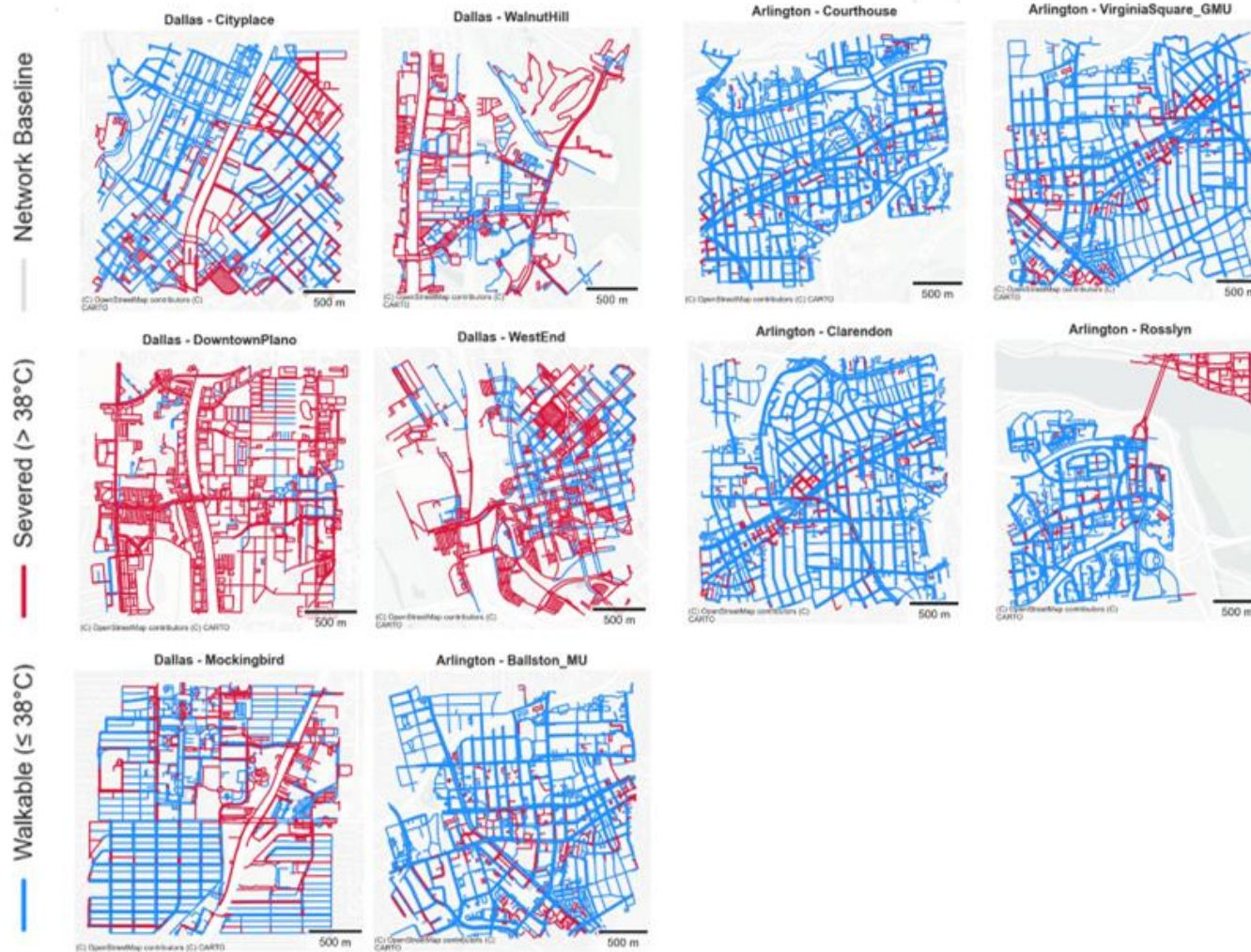
Figure 6. Representative severed-link patterns at peak conditions: Clarendon (Arlington) and Mockingbird (Dallas) under $\tau = 38^{\circ}\text{C}$ (summer)

Urban form plays a major role in determining whether the pedestrian network remains connected during extreme heat

More enclosed environments with shading tend to maintain connectivity, while more open environments experience much greater disruption

4. Exploratory Analysis: Results

Extreme Heat Produces Time-Specific Network Severance



Some station areas
experience **40–80% of links**
exceeding danger thresholds

Figure 7. Network severance analysis for all stations

4. Exploratory Analysis: Results

Magnitude and Duration of Network Disruption Vary by Station

Peak severance ranges from **<10% to nearly 90%** across stations

Some stations experience **multiple hours with severe disruption ($\geq 50\%$ severed)**

Others show only short, moderate disruption windows

Table 1 — Station-level severance timing and magnitude

City	Station	Onset hour ($T=38^{\circ}\text{C}$)	Peak hour ($T=38^{\circ}\text{C}$)	Peak $R_{sev}(\%)$ ($T=38^{\circ}\text{C}$)	Duration (h), $R_{sev}>0$ ($T=38^{\circ}\text{C}$)	Duration (h), $R_{sev}\geq 50\%$ ($T=38^{\circ}\text{C}$)
Arlington	Ballston	0	14	12.4	23	0
	Clarendon	12	14	6.9	4	0
	Courthouse	11	14	4.6	5	0
	Rosslyn	0	14	15.2	23	0
	Virginia Square	0	14	10.8	23	0
Dallas	Cityplace	0	13	45.9	24	0
	Downtown Plano	0	14	88.4	24	4
	Mockingbird	0	13	37.9	24	0
	WalnutHill	0	13	60.3	24	2
	WestEnd	0	13	62.5	24	4

4. Exploratory Analysis: Results

Transit Accessibility Loss Follows a Strong Diurnal Pattern

Accessibility loss rises through late morning

Peaks in mid-to-late afternoon

Declines in evening as thermal stress weakens

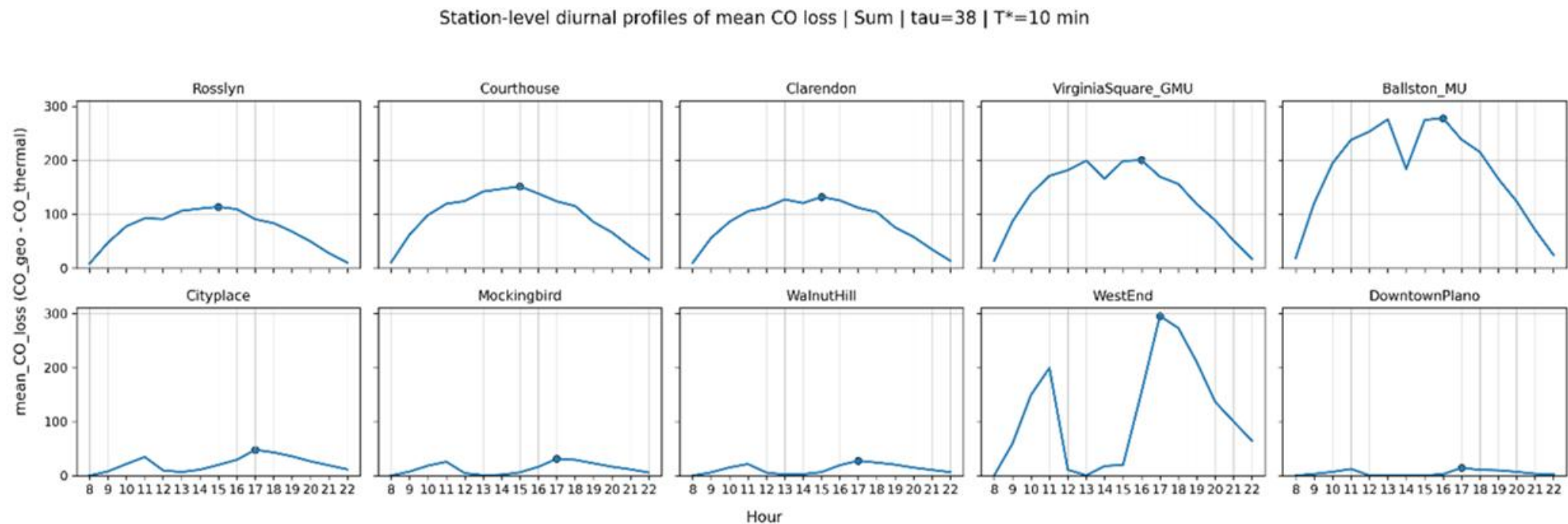


Figure 8. Diurnal profiles of cumulative opportunity loss

4. Exploratory Analysis: Results

Thermal Risk Exposure at Transit Stops (14:00)



Figure 9. Thermal Risk at Transit Stops Across Case Study Areas

4. Exploratory Analysis: Results

At Peak Heat, 35–41% of Reachable Transit Service Is Lost

Mean Accessibility Reduction Rate (ARR): **35–41%**

One-third to two-fifths of previously reachable service falls outside the 10-min threshold

Loss occurs **without any change in transit service**

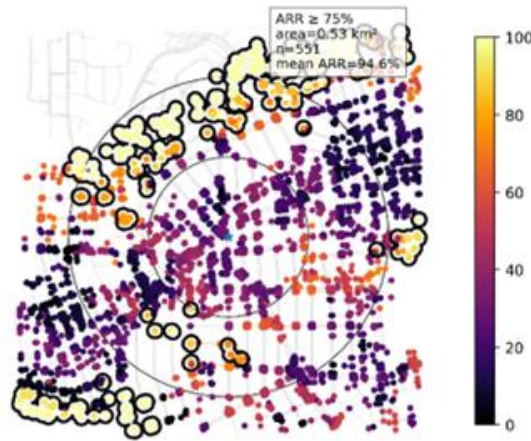
Table 2. Peak-hour ARR statistics

Station	Peak Hr	Coloss	Cogeo	Cotherm	ARRmean (%)	ARRmed (%)
Ballston	16	277.9	796.0	518.1	37.3	34.4
Virginia Square	16	200.2	610.4	410.2	35.5	32.3
Courthouse	15	151.5	431.0	279.5	41.0	31.3
Clarendon	15	132.1	392.7	260.6	38.6	31.0
Rosslyn	15	113.8	366.8	253.0	35.7	27.3
Cityplace	17	47.7	136.7	89.1	38.4	33.5
Mockingbird	17	31.1	96.2	65.1	40.9	35.5
Walnut Hill	17	27.5	80.1	52.6	39.3	28.6
Downtown Plano	17	14.4	49.3	34.9	35.2	27.3
West End	17	295.6	837.1	541.5	40.0	37.0

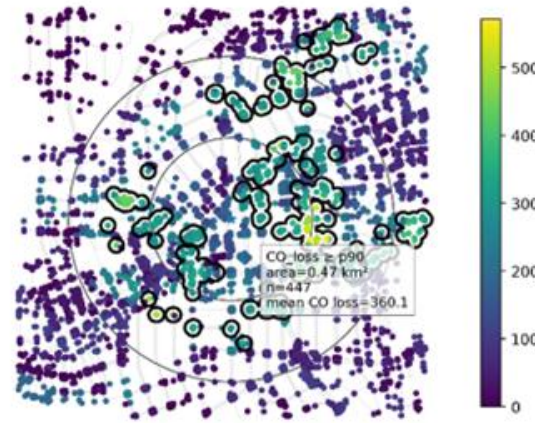
4. Exploratory Analysis: Results

Accessibility Loss Is Spatially Concentrated Along Key Corridors

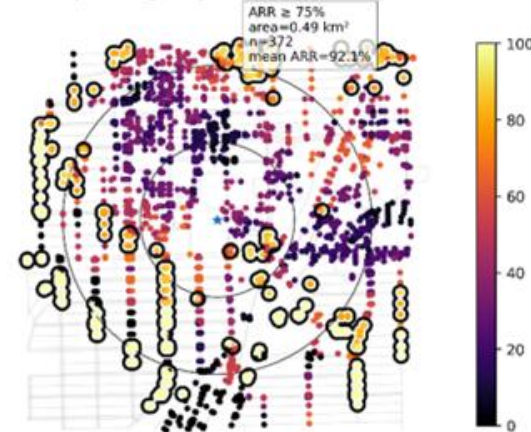
(a) ARR (%) | Clarendon | Sum h15 | $\tau=38.0$ | $T^*=10$ min



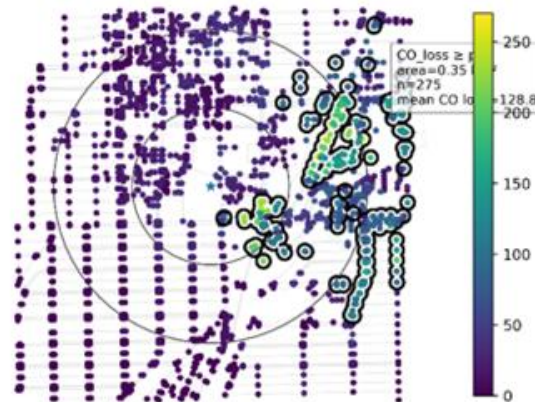
(b) CO loss (CO_{geo} – CO_{thermal}) | Clarendon | Sum h15



(a) ARR (%) | Mockingbird | Sum h17 | $\tau=38.0$ | $T^*=10$ min



(b) CO loss (CO_{geo} – CO_{thermal}) | Mockingbird | Sum h17



Severe accessibility reduction clusters along specific approach corridors

High-loss areas form **corridor-aligned bottlenecks**, not uniform patterns

Targeted improvements along these links could preserve large portions of access

Figure 10. Spatial distribution of ARR and CO loss

4. Exploratory Analysis: Results (Austin Case)

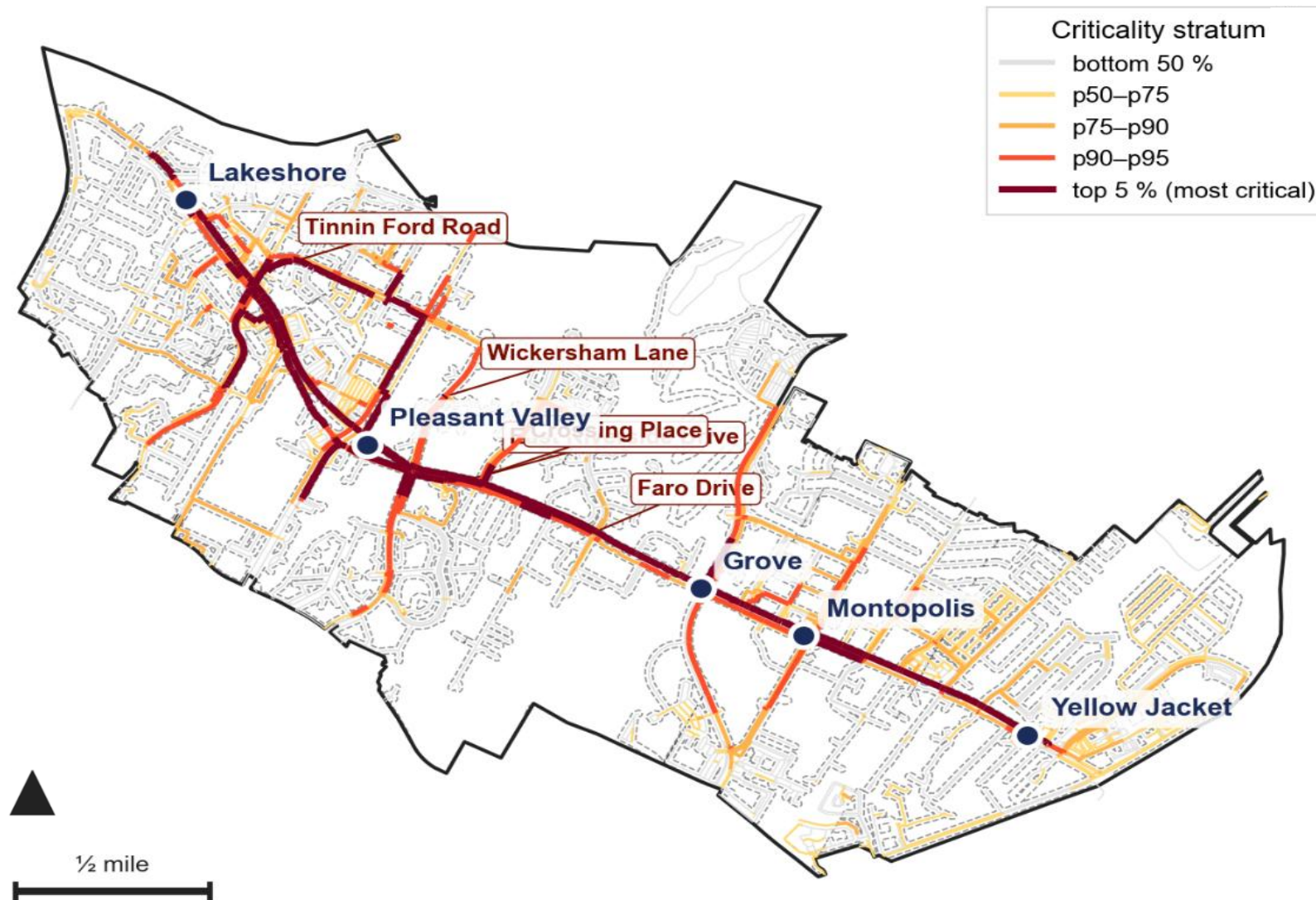


Figure 1. Street-segment heat criticality across the corridor pedestrian network (n = 4,115)

4. Exploratory Analysis: Results (Austin Case)

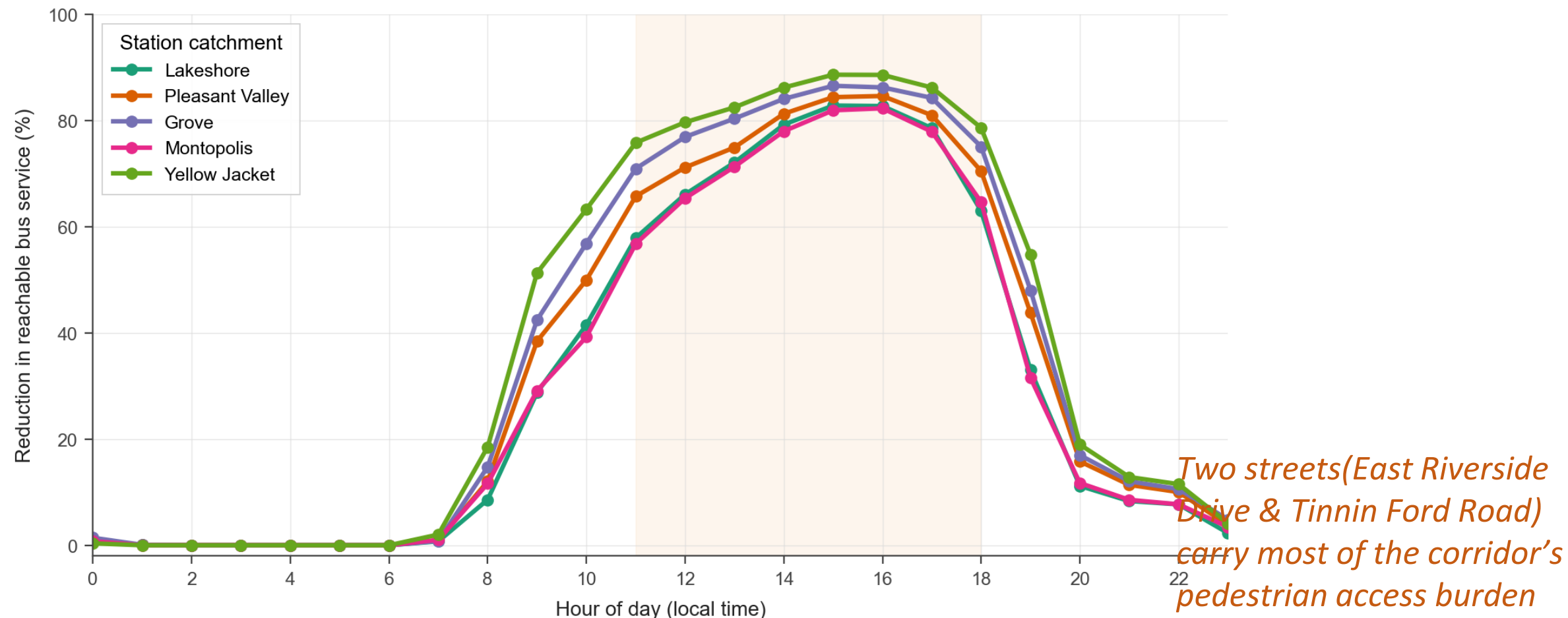


Figure 2. Diurnal variation in heat-adjusted transit accessibility reduction (ARR), 10-minute walk sheds.

Residents lose 80-90% of nearby bus service during summer afternoons

4. Exploratory Analysis: Results (Austin Case)

Pilot field work

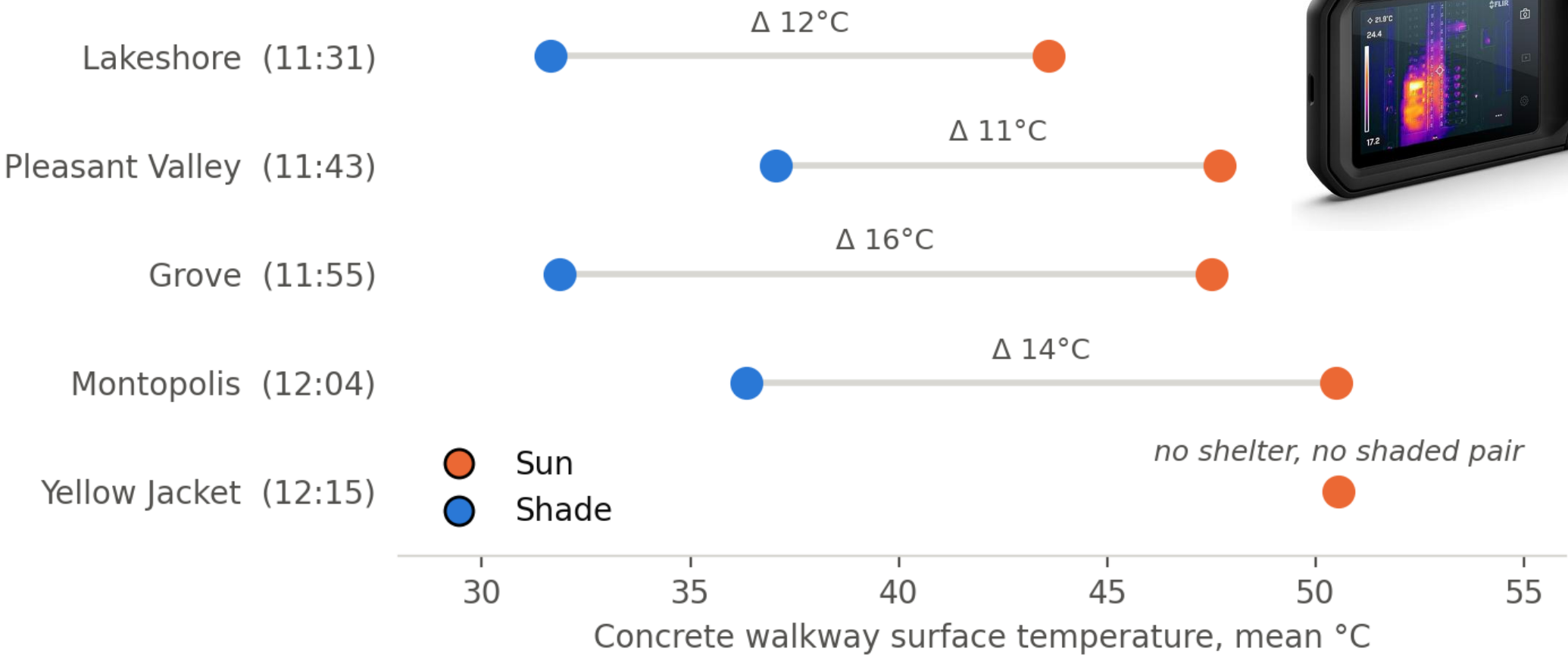


Figure 7. Paired sun–shade surface temperatures of identical materials at the five stops.

Shade is worth 11 to 16 °C; the only unsheltered bench reached 58 °C, a prolonged-contact burn hazard.

4. Exploratory Analysis: Results (Austin Case)

Pilot field work

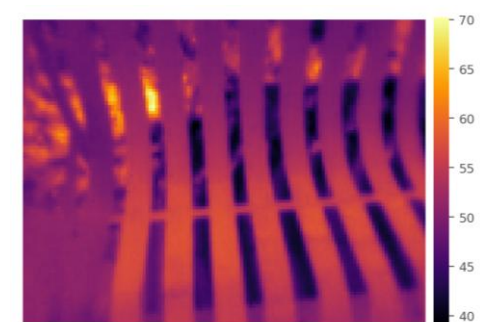
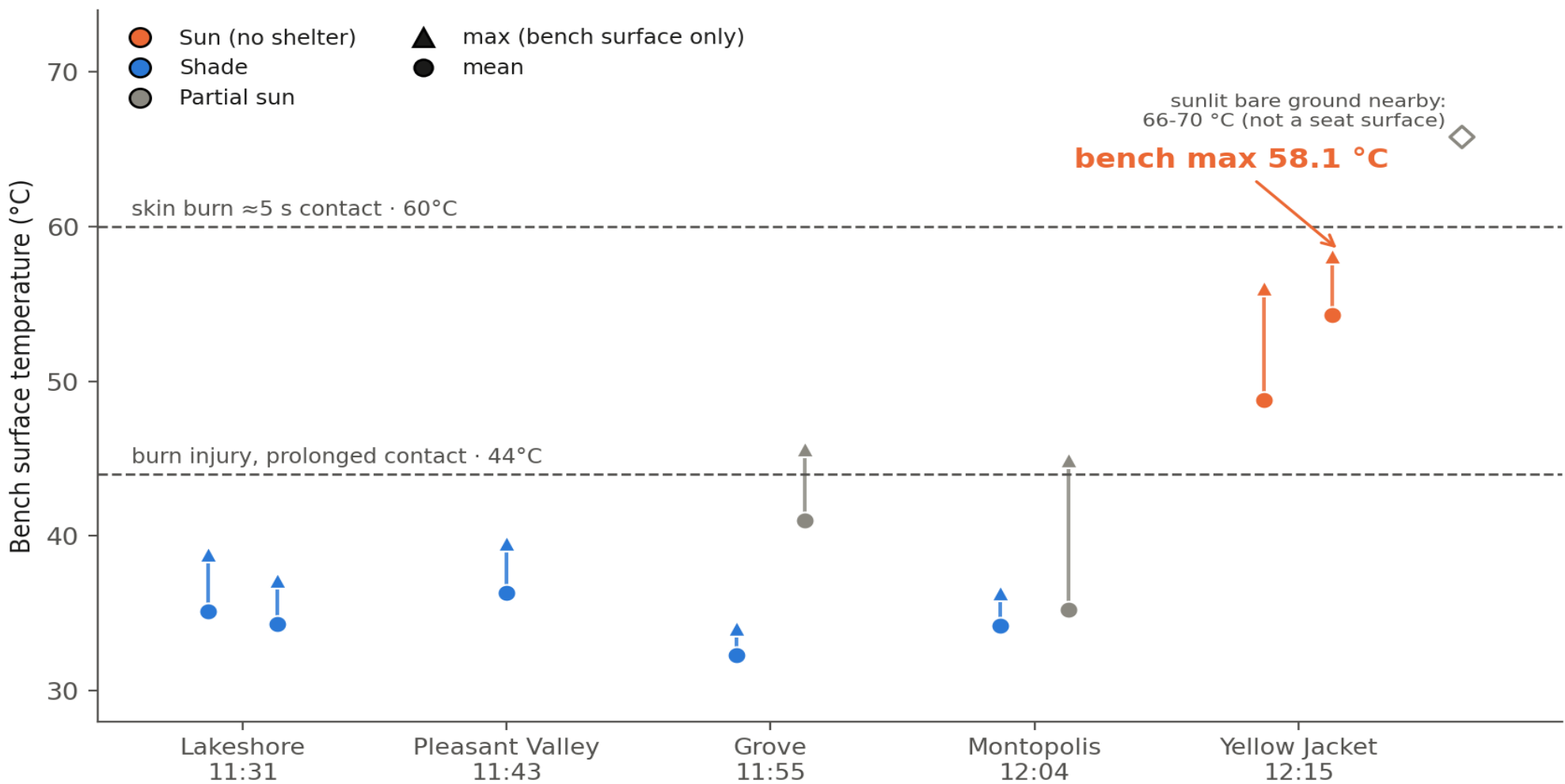


Figure 8. Maximum bench surface temperatures relative to the 60°C contact-burn threshold.

Figure 8a. 35°C in shade, 51°C in sun (Montopolis, 12:04 p.m.).

5. Concluding Remarks

Implications for Heat-Resilient Walkability for TOD:

Shade and thermal refuge should be treated as **mobility infrastructure**, not amenities

Prioritize **heat-critical pedestrian links** for intervention

Static walksheds **overestimate real access** under extreme heat

Apply **microclimate-informed design** at corridor and stop level

5. Concluding Remarks

Reconceptualize and operationalize walkability: From engineered to experienced

- Consider temporal microenvironmental dynamics
- Quantify thermal effects of built materials and forms to inform TOD planning and design
- Consider human-space (environment) interactions
- Assess experienced walkability by various demographic groups to achieve equitable TOD
- Applicable to many urban initiatives beyond TOD



Walkability: Foundational idea for 15-minute City

(Ravenscroft 2023. <https://www.dezeen.com/2023/10/16/15-minute-city-guide/>)

Acknowledgement

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